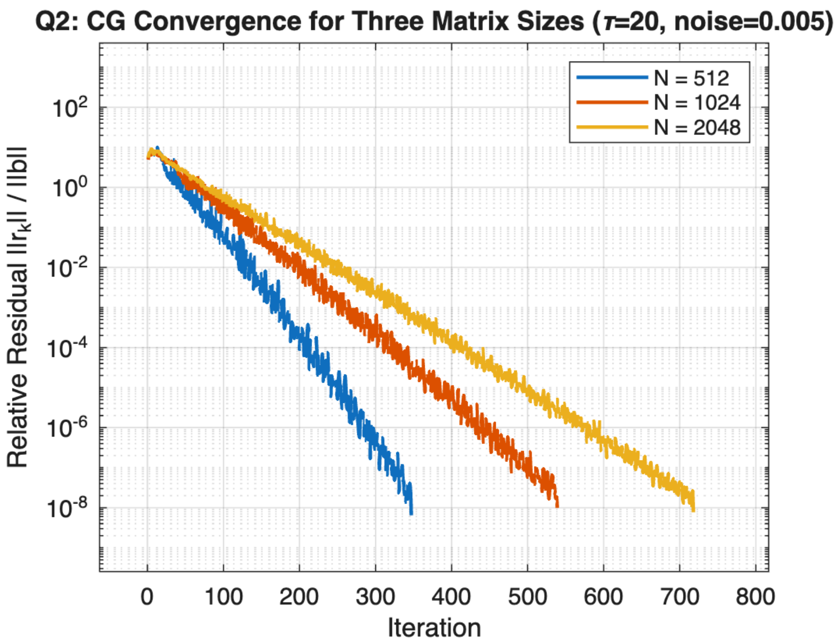
For this assignment, I implement the Conjugate Gradient method using MATLAB and the algorithm (6.18) from Saad second edition. The function is given by: *cg_solver (A, b, tol, max_iter)* and produces: the (approximate) solution, the number of iterations and the full history of the relative residuals (given by $||r_k|| / ||b||$) . To begin, an initial guess of *x = 0* is made (i.e. $x_o = 0$), then, the algorithm constructs *A-conjugate* search directions by making use of Fletcher-Reeves recurrence. If the relative residual drops below the specified tolerance, an early exit occurs. Similarly, if convergence does not occur, a warning is printed. For the second question, the CG algorithm is to be applied to the Matérn matrices with a specification of: $\tau = 20,\ noise = 0.005\ and\ tol = 10^{-8}$ across three different matrix dimensions. The results are given below:

N	Iterations	Final Rel. Residual	Time (s)
512	347	6.47×10^{-9}	0.01
1024	539	9.88×10^{-9}	0.03
2048	718	7.71×10^{-9}	0.20

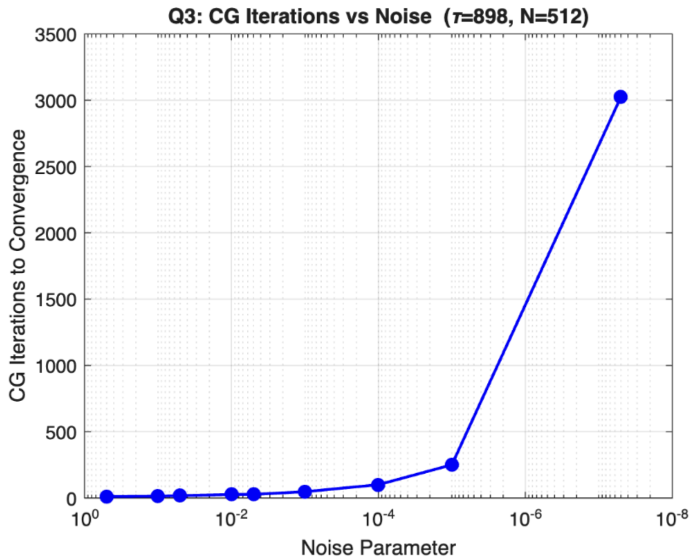


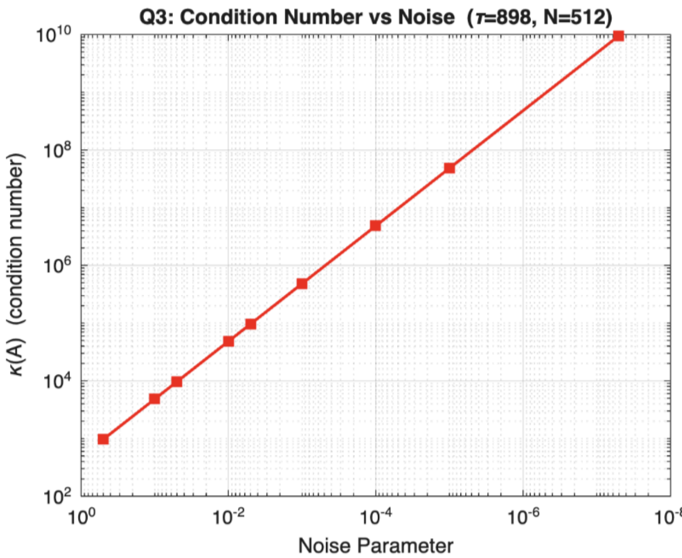
This behaviour is expected and consistent with the theoretical bounds of CG. While the convergence curves appear jagged, they are monotonically decreasing and are typical for CG on a dense correlated system. The condition number only grows slightly with a rise in N.

Following this, for question 3, I update the value of tau to 898 (the last three digits of student number) and range the noise parameter from $0.5 \rightarrow 5\,x\,10^{-8}$. The results are given below:

Noise	Iterations	Final Rel. Residual	κ(A)
5.00×10^{-1}	11	7.49×10^{-10}	9.68×10^{2}
1.00×10^{-1}	14	5.65×10^{-9}	4.84×10^{3}
5.00×10^{-2}	18	6.60×10^{-10}	9.67×10^{3}
1.00×10^{-2}	27	5.26×10^{-10}	4.84×10^{4}
5.00×10^{-3}	28	4.89×10^{-9}	9.67×10^{4}
1.00×10^{-3}	48	6.57×10^{-9}	4.84×10^{5}
1.00×10^{-4}	101	7.90×10^{-9}	4.84×10^{6}
1.00×10^{-5}	251	9.39×10^{-9}	4.84×10^{7}
5.00×10^{-8}	3025	9.88×10^{-9}	9.56×10^{9}

This time, the noise term shifts all of the eigenvalues up by an amount equal to noise. The condition number is scaling inversely to the noise across the entire range. As noise decreases, the condition number grows and as such, the iteration count is expected to grow. This trend becomes sharper when dropping below noise of 10^{-5}, there is a loss of numerical orthogonality amongst the conjugate search directions and means the CG algorithms (re)traverses directions that it had already minimised. This occurs due to the accumulation of rounding errors via each matrix-vector product operation and increases the number of iterations.





For the fourth question, the preconditioned system is given by: *M =chol(a); A_new = M \ (A / M');* where M is the upper Cholesky factor. This helps to transform the system to $M^{-1}AM^{-T}y = M^{-1}b$, with the solution being: $x = M^{-T}y$

The results of this are given below:

System	Iterations	Final Rel. Residual	κ(matrix)
A (unpreconditioned)	112	7.45×10^{-9}	4.40×10^{3}
A_new (Cholesky precond)	12	3.39×10^{-10}	4.58×10^{4}

From the above, we can see that the Cholesky preconditioning helps A_new converge in 12 iterations versus the A with no preconditioning which converges in 112, that's roughly a ~9x speedup. Moreover, the Cholesky version does this with a higher condition number. The convergence of the CG is influenced by clustering of eigenvalues too, not just the condition number. In the Cholesky implementation, the factorisation introduces rounding and helps the eigenvalues to cluster around 1 with not as many outliers. This convergence difference can be seen below: